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**GISMA UNIVERSITY OF APPLIED SCIENCES SCHOOL OF
COMPUTER SCIENCE**

**B104 ARTIFICIAL INTELLIGENCE & MACHINE
LEARNING**

TITLE: INDIVIDUAL PROJECT

Executive Summary

The purpose of this report is to perform some data analysis over the Bank Marketing data set to find some pattern or clue that shows the factors concerning term-deposit subscription. Chi square, ANOVA, wilcoxon, correlation and logistic regression methods were used. The results show that there are significant relationships among loans, the type of campaign elected and call length. The results indicate that better customer targeting can be obtained, however the class imbalance and low value of the model sensitivity constrain the predictive value.

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1. Introduction and Business Problem

1.1 Business Problem

Banks are spending substantial resources on direct marketing, but a comparatively low percentage of the customers contacted put money into term deposits. Utilizing customer characteristics and related data that is important to the campaign to understand the relationship of these attributes with the customer outcomes will significantly help optimize marketing efforts, together with upcoming action.

1.2 Business Questions

1. What are the important characteristics of customers which are significantly related to term-deposit subscription?
2. What is the relationship between the factors of past and present marketing campaigns and customers' subscription?
3. What might influence the probability that a customer will invest in a term deposit?

1.3 Objectives

- To explore customers' demographic data, their financial data, campaign data and term-deposit subscription patterns, using descriptive statistics and visualisations.
- To identify statistically inferential statistical methods to determine whether there is a statistically significant relationship between customer/campaign characteristics and term-deposit subscription.
- To recommend for data-driven approaches that might help the bank sharpen their customer targeting and make their term-deposit marketing more effective in the future.

2. Dataset Description

2.1 Data Source

The data source for this analysis is the Bank Marketing dataset, which can be found in UCI Machine Learning Repository (UCI, 2026). It comprises data from direct marketing campaigns carried out by a Portuguese banking institution, data related to the characteristics of potential customers, financial characteristics, data on the characteristics of the campaigns, and term-deposit subscription's outcomes.

2.2 Variables

```

Console Terminal Background Jobs
R - R 4.5.1 . ~/
> library(readr)
> bank <- read_csv("C:/users/user/downloads/bank+marketing/bank.csv")
ROWS: 4521 Columns: 17
— Column specification —
Delimiter: ","
chr (10): job, marital, education, default, housing, loan, contact, month, poutcome, y
dbl (7): age, balance, day, duration, campaign, pdays, previous

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
> view(bank)
> head(bank)
# A tibble: 6 x 17
  age job marital education default balance housing loan contact day month duration campaign pdays previous
  <dbl> <chr> <chr> <chr> <chr> <dbl> <chr> <chr> <chr> <dbl> <chr> <dbl> <dbl> <dbl> <dbl>
1 30 unemplo... married primary no 1787 no no cellul... 19 oct 79 1 -1 0
2 33 services married secondary no 4789 yes yes cellul... 11 may 220 1 339 4
3 35 managem... single tertiary no 1350 yes no cellul... 16 apr 185 1 330 1
4 30 managem... married tertiary no 1476 yes yes unknown 3 jun 199 4 -1 0
5 59 blue-co... married secondary no 0 yes no unknown 5 may 226 1 -1 0
6 35 managem... single tertiary no 747 no no cellul... 23 feb 141 2 176 3
# i 2 more variables: poutcome <chr>, y <chr>
> tail(bank)
# A tibble: 6 x 17
  age job marital education default balance housing loan contact day month duration campaign pdays previous
  <dbl> <chr> <chr> <chr> <chr> <dbl> <chr> <chr> <chr> <dbl> <chr> <dbl> <dbl> <dbl>
1 32 services single secondary no 473 yes no cellul... 7 jul 624 5 -1 0
2 33 services married secondary no -333 yes no cellul... 30 jul 329 5 -1 0
3 57 self-em... married tertiary yes -313 yes yes unknown 9 may 153 1 -1 0
4 57 technic... married secondary no 295 no no cellul... 19 aug 151 11 -1 0
5 28 blue-co... married secondary no 1137 no no cellul... 6 feb 129 4 211 3
6 44 entrepr... single tertiary no 1136 yes yes cellul... 3 apr 345 2 249 7
# i 2 more variables: poutcome <chr>, y <chr>
> dim(bank)
[1] 4521 17
> names(bank)
[1] "age" "job" "marital" "education" "default" "balance" "housing" "loan" "contact"
[10] "day" "month" "duration" "campaign" "pdays" "previous" "poutcome" "y"

```

Figure 1: Structure and Variables of the Bank Marketing Dataset

There are a total of 4521 observations and 17 variables. Age, job, marital, education, default, balance, housing, loan, contact, day, month, duration, campaign, pdays, previous, poutcome, and y are the variables to be considered as shown in Figure 1. These include customer details, monetary data, campaign details, and the conclusion of the subscription.

3. Data Preparation, Sampling and Cleaning

```

> dim(bank)
[1] 4521 17
> names(bank)
[1] "age"      "job"      "marital"  "education" "default"  "balance"  "housing"  "loan"      "contact"
[10] "day"      "month"    "duration" "campaign"  "pdays"   "previous" "poutcome" "y"
> str(bank)
spec_tbl_ [4,521 x 17] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ age      : num [1:4521] 30 33 35 30 59 35 36 39 41 43 ...
 $ job      : chr [1:4521] "unemployed" "services" "management" "management" ...
 $ marital  : chr [1:4521] "married" "married" "single" "married" ...
 $ education: chr [1:4521] "primary" "secondary" "tertiary" "tertiary" ...
 $ default  : chr [1:4521] "no" "no" "no" "no" ...
 $ balance  : num [1:4521] 1787 4789 1350 1476 0 ...
 $ housing  : chr [1:4521] "no" "yes" "yes" "yes" ...
 $ loan     : chr [1:4521] "no" "yes" "no" "yes" ...
 $ contact  : chr [1:4521] "cellular" "cellular" "cellular" "unknown" ...
 $ day      : num [1:4521] 19 11 16 3 5 23 14 6 14 17 ...
 $ month    : chr [1:4521] "oct" "may" "apr" "jun" ...
 $ duration : num [1:4521] 79 220 185 199 226 141 341 151 57 313 ...
 $ campaign : num [1:4521] 1 1 1 4 1 2 1 2 2 1 ...
 $ pdays    : num [1:4521] -1 339 330 -1 -1 176 330 -1 -1 147 ...
 $ previous : num [1:4521] 0 4 1 0 0 3 2 0 0 2 ...
 $ poutcome : chr [1:4521] "unknown" "failure" "failure" "unknown" ...
 $ y        : chr [1:4521] "no" "no" "no" "no" ...
- attr(*, "spec")=
.. cols(
..   age = col_double(),
..   job = col_character(),
..   marital = col_character(),
..   education = col_character(),
..   default = col_character(),
..   balance = col_double(),
..   housing = col_character(),
..   loan = col_character(),
..   contact = col_character(),
..   day = col_double(),
..   month = col_character(),
..   duration = col_double(),
..   campaign = col_double(),
..   pdays = col_double(),
..   previous = col_double(),
..   poutcome = col_character(),
..   y = col_character()
.. )
- attr(*, "problems")=<externalptr>

```

Figure 2: Dataset Structure and Variable Types

4,521 observations of 17 variables (including numeric and non-numeric variables). These are variables like customer's demographics, financial characteristics, information about the campaign and y will indicate whether a customer has subscribed to the term deposit.

```

> summary(bank)
  age      job      marital      education      default      balance
Min.   :19.00 Length:4521 Length:4521 Length:4521 Length:4521 Min.   :-3313
1st Qu.:33.00 Class :character Class :character Class :character Class :character 1st Qu.: 69
Median :39.00 Mode  :character Mode  :character Mode  :character Mode  :character Median : 444
Mean   :41.17                                     Mean   :1423
3rd Qu.:49.00                                     3rd Qu.:1480
Max.   :87.00                                     Max.   :71188

  housing      loan      contact      day      month      duration
Length:4521 Length:4521 Length:4521 Min.   : 1.00 Length:4521 Min.   : 4
Class :character Class :character Class :character 1st Qu.: 9.00 Class :character 1st Qu.:104
Mode  :character Mode  :character Mode  :character Median :16.00 Mode  :character Median :185
                                     Mean   :15.92                                     Mean   :264
                                     3rd Qu.:21.00                                     3rd Qu.:329
                                     Max.   :31.00                                     Max.   :3025

  campaign      pdays      previous      poutcome      y
Min.   : 1.000 Min.   : -1.00 Min.   : 0.0000 Length:4521 Length:4521
1st Qu.: 1.000 1st Qu.: -1.00 1st Qu.: 0.0000 Class :character Class :character
Median : 2.000 Median : -1.00 Median : 0.0000 Mode  :character Mode  :character
Mean   : 2.794 Mean   :39.77 Mean   : 0.5426                                     Mean   :
3rd Qu.: 3.000 3rd Qu.: -1.00 3rd Qu.: 0.0000                                     3rd Qu.:
Max.   :50.000 Max.   :871.00 Max.   :25.0000                                     Max.   :

```

Figure 3: Summary Statistics of the Bank Marketing Dataset

Numerical variables exhibit quite a bit of variation, as seen in the descriptive summary. The average age of the customer is 41.17 with balance average being 1423.

```

> # 2.CHECK DATA QUALITY
> # Check missing values
> sum(is.na(bank))
[1] 0
> colSums(is.na(bank))
  age      job      marital education  default  balance  housing    loan  contact    day    month  duration  campaign  pdays  previous
0     0       0           0         0        0        0        0       0       0       0       0        0        0        0        0
poutcome
0     0
> # Check duplicate observations
> sum(duplicated(bank))
[1] 0
> # Display duplicates if any
> bank[duplicated(bank), ]
# A tibble: 0 x 17
# i 17 variables: age <dbl>, job <chr>, marital <chr>, education <chr>, default <chr>, balance <dbl>, housing <chr>, loan <chr>, contact <chr>,
#   day <dbl>, month <chr>, duration <dbl>, campaign <dbl>, pdays <dbl>, previous <dbl>, poutcome <chr>, y <chr>
> # Check blank values in character variables
> sapply(bank, function(x) sum(x == "", na.rm = TRUE))
  age      job      marital education  default  balance  housing    loan  contact    day    month  duration  campaign  pdays  previous
0     0       0           0         0        0        0        0       0       0       0       0        0        0        0        0
poutcome
0     0

```

Figure 4: Missing Values, Duplicates and Blank-Value Assessment

There are zero missing values, zero duplicated observations, and zero blank values of all 17 variables, as noted in the data-quality assessment. Thus, nothing has to be removed, just because of traditional missing-, duplicate- or blank data.

```

> # 3. CHECK CATEGORICAL VALUES
> table(bank$job)
  admin.  blue-collar  entrepreneur  housemaid  management  retired self-employed  services  student  technician
    478         946         168         112         969         230         183         417         84         768
 unemployed  unknown
    128         38
> table(bank$marital)
divorced  married  single
    528    2797    1196
> table(bank$education)
  primary secondary tertiary  unknown
    678     2306     1350     187
> table(bank$default)
no  yes
4445 76
> table(bank$housing)
no  yes
1962 2559
> table(bank$loan)
no  yes
3830 691
> table(bank$contact)
cellular telephone  unknown
    2896         301     1324
> table(bank$month)
apr  aug  dec  feb  jan  jul  jun  mar  may  nov  oct  sep
293  633   20  222  148  706  531   49 1398  389   80   52
> table(bank$poutcome)
failure  other success unknown
    490    197    129   3705
> table(bank$y)
no  yes
4000 521
> |

```

Figure 5: Frequency Distribution of Categorical Variables

Several 'unknown' values are found in the categorical inspection - most prominently poutcome (3,705), contact (1,324), education (187) and job (38).


```

> # 5. CONVERT CATEGORICAL VARIABLES TO FACTORS
> factor_variables <- c("job", "marital", "education", "default",
+ "housing", "loan", "contact", "month",
+ "poutcome", "y")
> bank[factor_variables] <- lapply(bank[factor_variables], factor)
> str(bank)
spc_tbl_ [4,521 × 17] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ age      : num [1:4521] 30 33 35 30 59 35 36 39 41 43 ...
 $ job      : Factor w/ 12 levels "admin.", "blue-collar",...: 11 8 5 5 2 5 7 10 3 8 ...
 $ marital  : Factor w/ 3 levels "divorced", "married",...: 2 2 3 2 2 3 2 2 2 2 ...
 $ education: Factor w/ 4 levels "primary", "secondary",...: 1 2 3 3 2 3 3 2 3 1 ...
 $ default  : Factor w/ 2 levels "no", "yes": 1 1 1 1 1 1 1 1 1 1 ...
 $ balance  : num [1:4521] 1787 4789 1350 1476 0 ...
 $ housing  : Factor w/ 2 levels "no", "yes": 1 2 2 2 2 1 2 2 2 2 ...
 $ loan     : Factor w/ 2 levels "no", "yes": 1 2 1 2 1 1 1 1 1 2 ...
 $ contact  : Factor w/ 3 levels "cellular", "telephone",...: 1 1 1 3 3 1 1 1 3 1 ...
 $ day      : num [1:4521] 19 11 16 3 5 23 14 6 14 17 ...
 $ month    : Factor w/ 12 levels "apr", "aug", "dec",...: 11 9 1 7 9 4 9 9 9 1 ...
 $ duration : num [1:4521] 79 220 185 199 226 141 341 151 57 313 ...
 $ campaign : num [1:4521] 1 1 1 4 1 2 1 2 2 1 ...
 $ pdays    : num [1:4521] -1 339 330 -1 -1 176 330 -1 -1 147 ...
 $ previous : num [1:4521] 0 4 1 0 0 3 2 0 0 2 ...
 $ poutcome : Factor w/ 4 levels "failure", "other",...: 4 1 1 4 4 1 2 4 4 1 ...
 $ y        : Factor w/ 2 levels "no", "yes": 1 1 1 1 1 1 1 1 1 1 ...
- attr(*, "spec")=
.. cols(
..   age = col_double(),
..   job = col_character(),
..   marital = col_character(),
..   education = col_character(),
..   default = col_character(),
..   balance = col_double(),
..   housing = col_character(),
..   loan = col_character(),
..   contact = col_character(),
..   day = col_double(),
..   month = col_character(),
..   duration = col_double(),
..   campaign = col_double(),
..   pdays = col_double(),
..   previous = col_double(),
..   poutcome = col_character(),
..   y = col_character()
.. )
- attr(*, "problems")=<externalptr>

```

Figure 6: Conversion of Categorical Variables to Factors

The data for categorical variables were coded in R from characters to factors for the purpose of appropriate statistical analysis. The data set was subjected to a data structure, data types, missing values, duplicates and categorical distributions. In R, exploratory and inferential statistical analysis was done on the relevant categorical variables which were recoded as factors (Finch, 2025). There were no missed observations, no blank observations and there was no duplication of observations. But there were also several categorical variables that had “unknown” values in their attributes.

4. Exploratory Data Analysis and Descriptive Statistics

4.1 Descriptive Statistics

```

> # 6. DESCRIPTIVE STATISTICS
> # Overall summary
> summary(bank)
  age                job                marital      education  default    balance    housing    loan    contact    day
Min.   :19.00   management :969   divorced: 528   primary   : 678   no :4445   Min.   : -3313   no :1962   no :3830   cellular :2896   Min.   : 1.00
1st Qu.:33.00   blue-collar:946   married :2797   secondary:2306   yes:  76   1st Qu.:  69   1st Qu.:  69   yes:2559   yes: 691   telephone:301   1st Qu.: 9.00
Median :39.00   technician :768   single  :1196   tertiary :1350   3rd Qu.:1423   Median : 444   3rd Qu.:1480   3rd Qu.:1480   unknown :1324   Median :16.00
Mean   :41.17   admin.    :478   unknown  :187   Max.   :71188   Mean   : 1423   Max.   :71188
3rd Qu.:49.00   services  :417   retired  :230   (other)  :713
Max.   :87.00
  month    duration    campaign    pdays    previous    poutcome    y
may :1398   Min.   : 4   Min.   : 1.000   Min.   : -1.00   Min.   : 0.0000   failure: 490   no :4000
jul  : 706   1st Qu.:104   1st Qu.: 1.000   1st Qu.: -1.00   1st Qu.: 0.0000   other : 197   yes: 521
aug  : 633   Median :185   Median : 2.000   Median : -1.00   Median : 0.0000   success:129
jun  : 531   Mean   :264   Mean   : 2.794   Mean   : 39.77   Mean   : 0.5426   unknown:3705
nov  : 389   3rd Qu.:329   3rd Qu.: 3.000   3rd Qu.: -1.00   3rd Qu.: 0.0000
apr  : 293   Max.   :3025   Max.   :50.000   Max.   :871.00   Max.   :25.0000
  (other): 571
> # Numerical variables
> numeric_variables <- c("age", "balance", "day", "duration",
+ "campaign", "pdays", "previous")
> summary(bank[numeric_variables])
  age                balance            day            duration    campaign    pdays    previous
Min.   :19.00   Min.   : -3313   Min.   : 1.00   Min.   : 4   Min.   : 1.000   Min.   : -1.00   Min.   : 0.0000
1st Qu.:33.00   1st Qu.:  69   1st Qu.: 9.00   1st Qu.:104   1st Qu.: 1.000   1st Qu.: -1.00   1st Qu.: 0.0000
Median :39.00   Median : 444   Median :16.00   Median :185   Median : 2.000   Median : -1.00   Median : 0.0000
Mean   :41.17   Mean   :1423   Mean :15.92   Mean   :264   Mean   : 2.794   Mean   : 39.77   Mean   : 0.5426
3rd Qu.:49.00   3rd Qu.:1480   3rd Qu.:21.00   3rd Qu.:329   3rd Qu.: 3.000   3rd Qu.: -1.00   3rd Qu.: 0.0000
Max.   :87.00   Max.   :71188   Max.   :31.00   Max.   :3025   Max.   :50.000   Max.   :871.00   Max.   :25.0000

```

Figure 7: Descriptive Statistics of the Bank Marketing Dataset

The descriptive statistics indicate significant differences between customer and campaign attributes. Customer's average age is about 41 years, and the balance and call duration have a broad range.

4.2 Data Visualisation

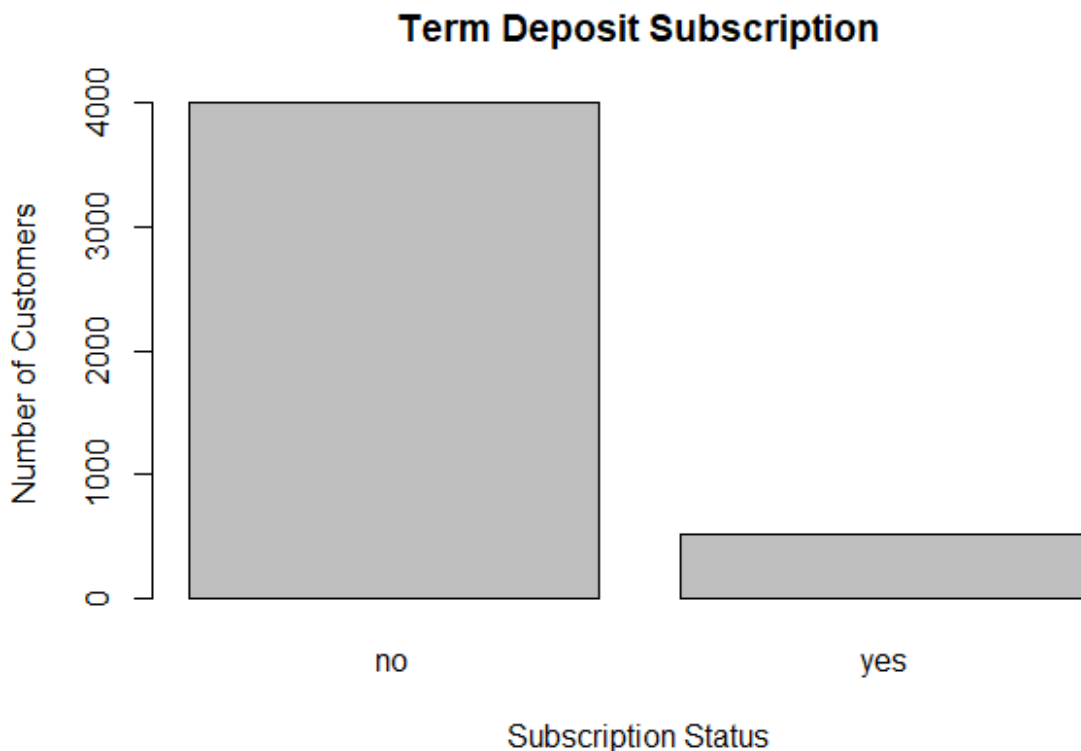


Figure 8: Distribution of Term Deposit Subscription

This relationship is well skewed in favor of the subscription outcome, as 400 customers are not subscribed in relation to 521 subscribers.

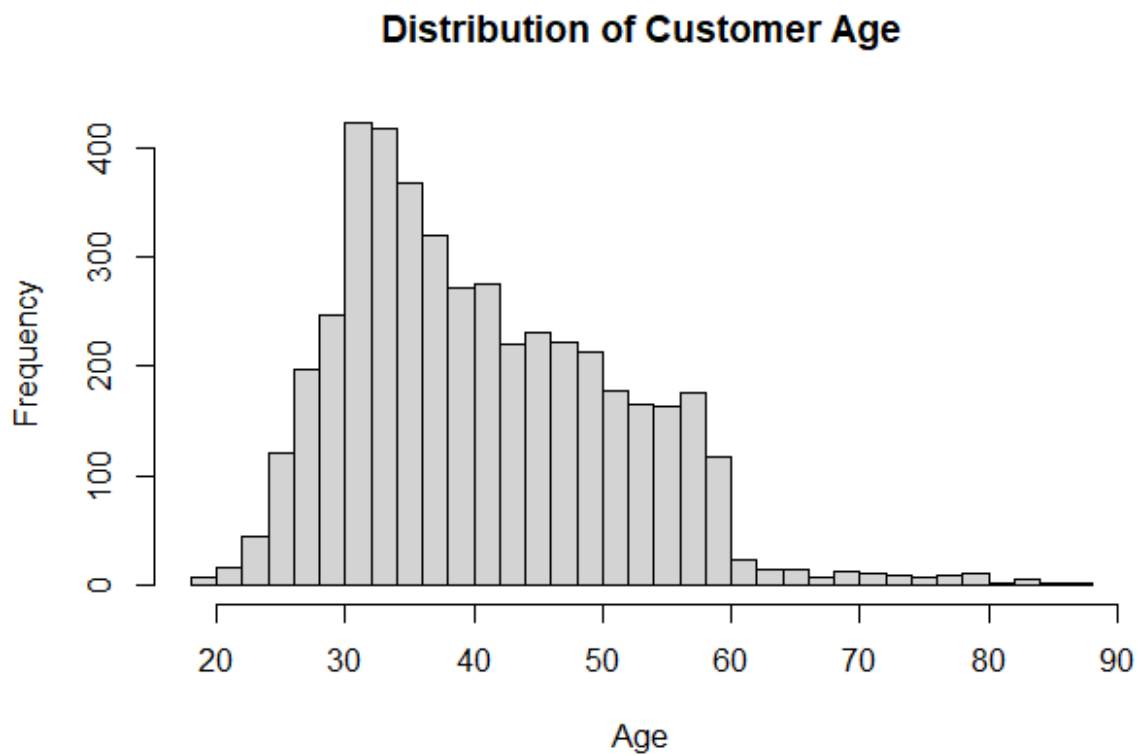


Figure 9: Distribution of Customer Age

The age of the customers varies between 19 and 87 years, but most of the observations have been done between the age of 30 to 60 years.

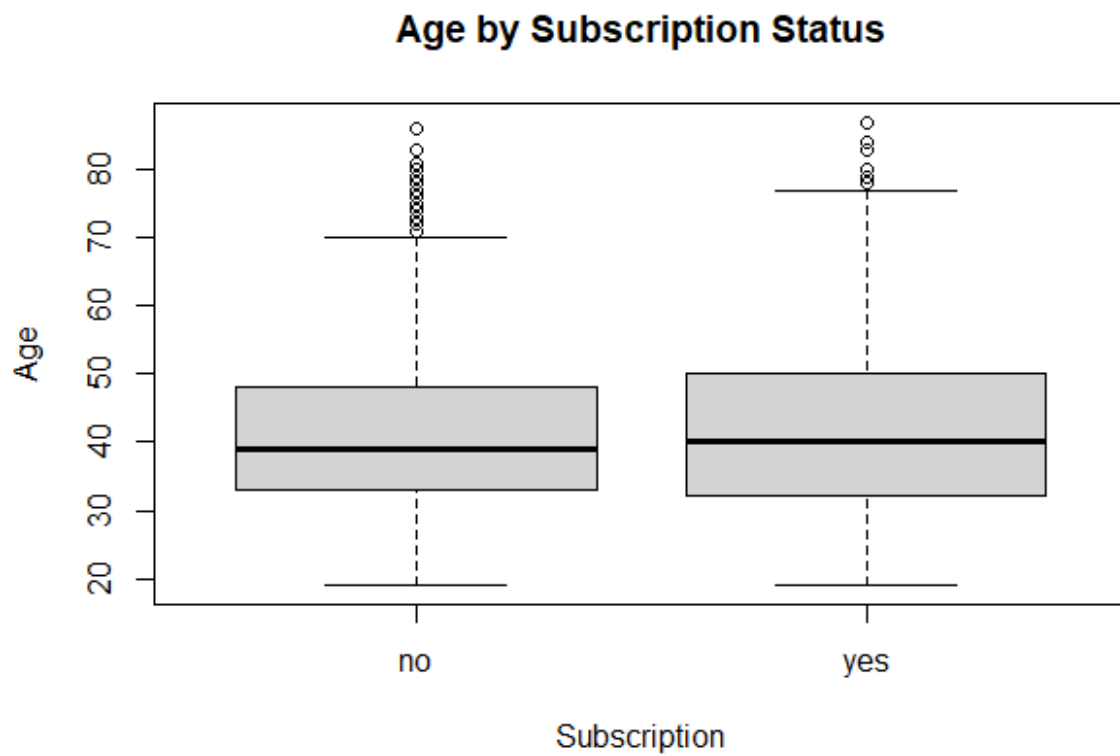


Figure 10: Customer Age by Subscription Status

The age distribution for subscribers and non-subscribers are very similar and have median ages of nearly 40 years.

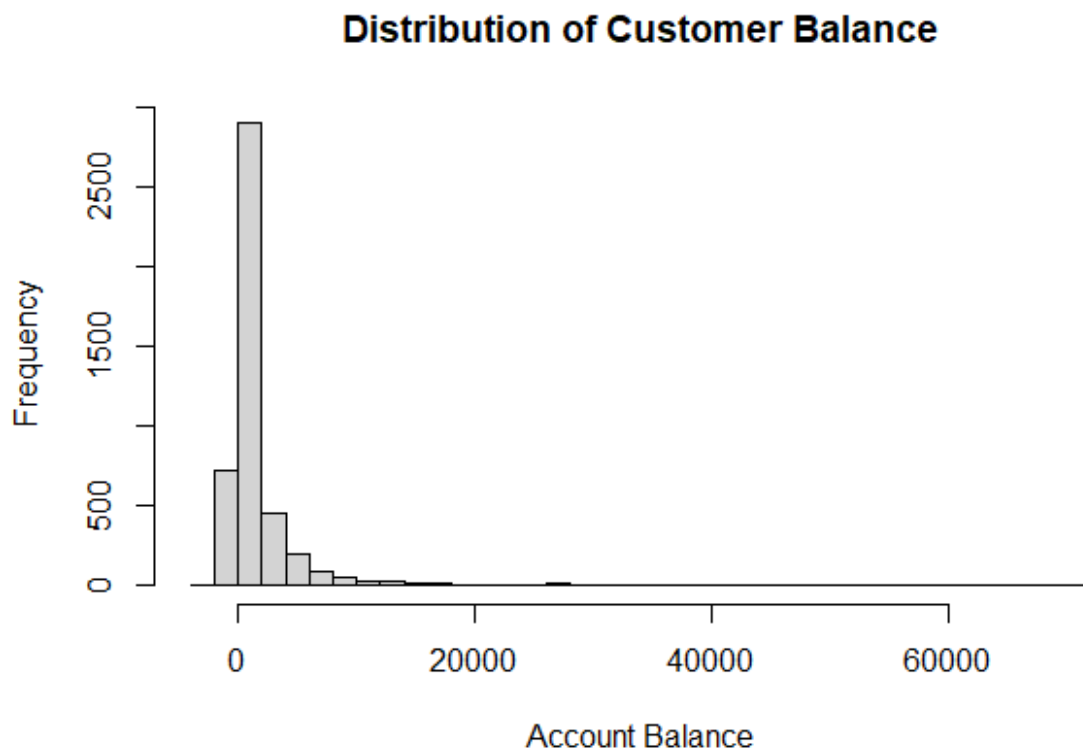


Figure 11: Distribution of Customer Account Balance

The reason is that customers' account balance is highly right-skewed, and thus susceptible to some relatively low account balances and a few very high ones.

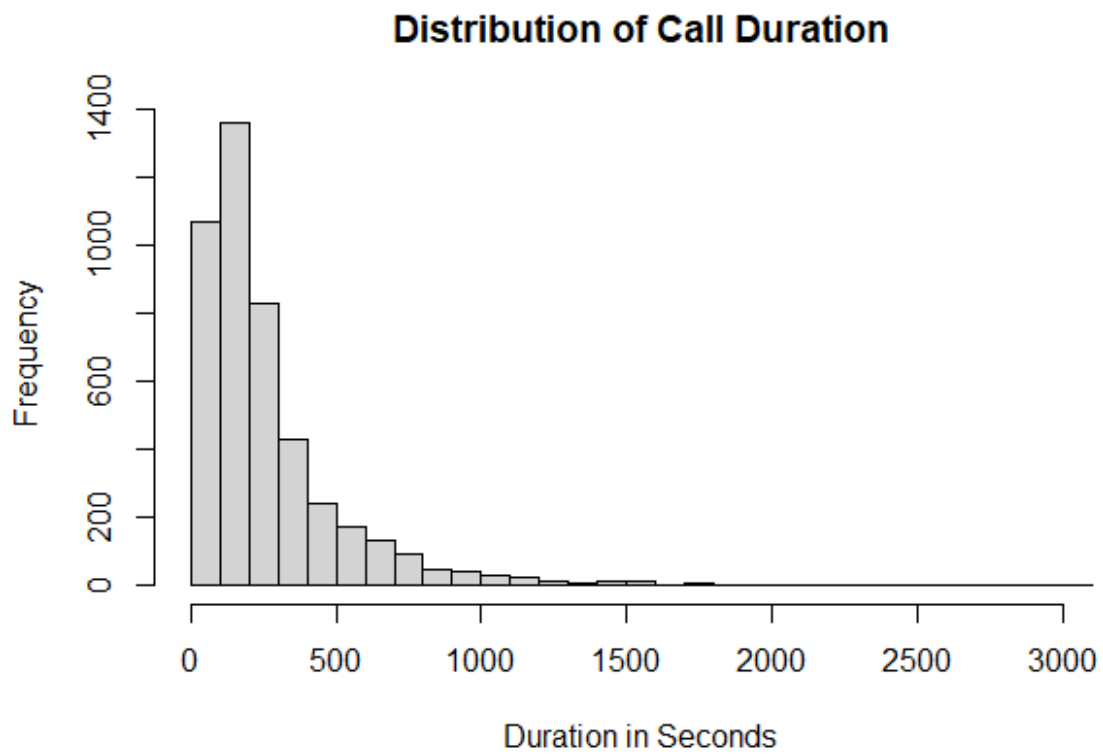


Figure 12: Distribution of Call Duration

The Call Duration when plotted is highly right skewed. The sizes of the customer calls generally got shorter and a gradually smaller proportion of calls were very long, to the order of ~3,025 seconds.

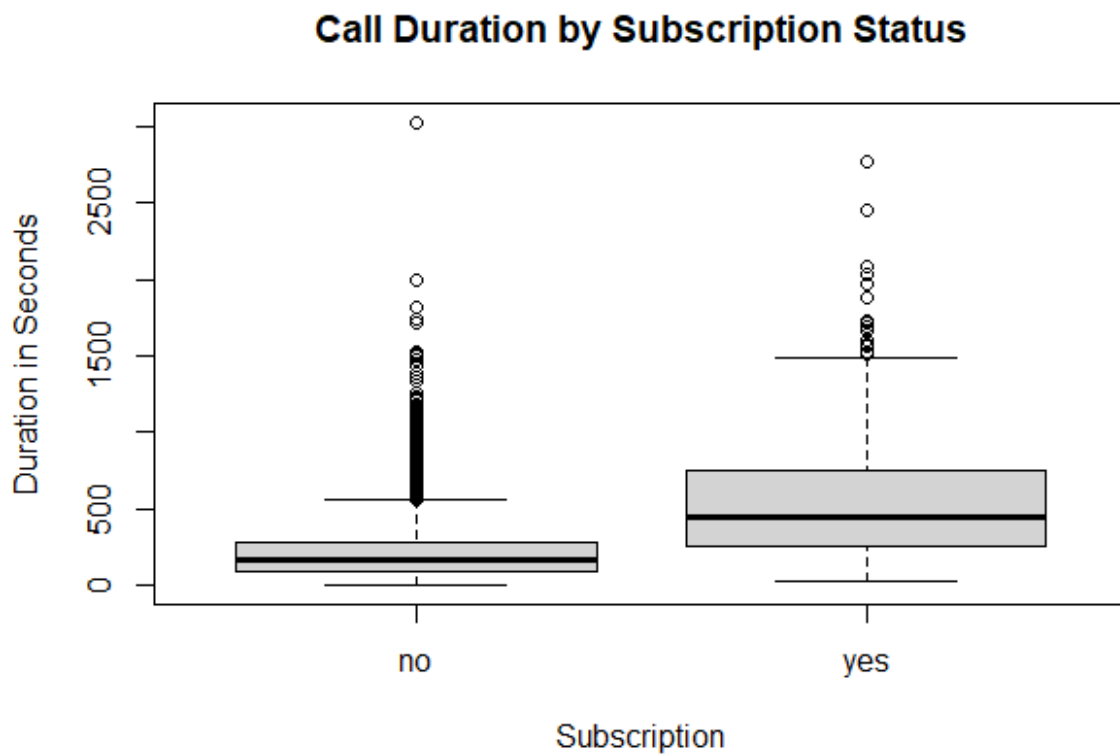


Figure 13: Call Duration by Subscription Status

Customers that subscribed had generally longer calls than customers who didn't subscribe.

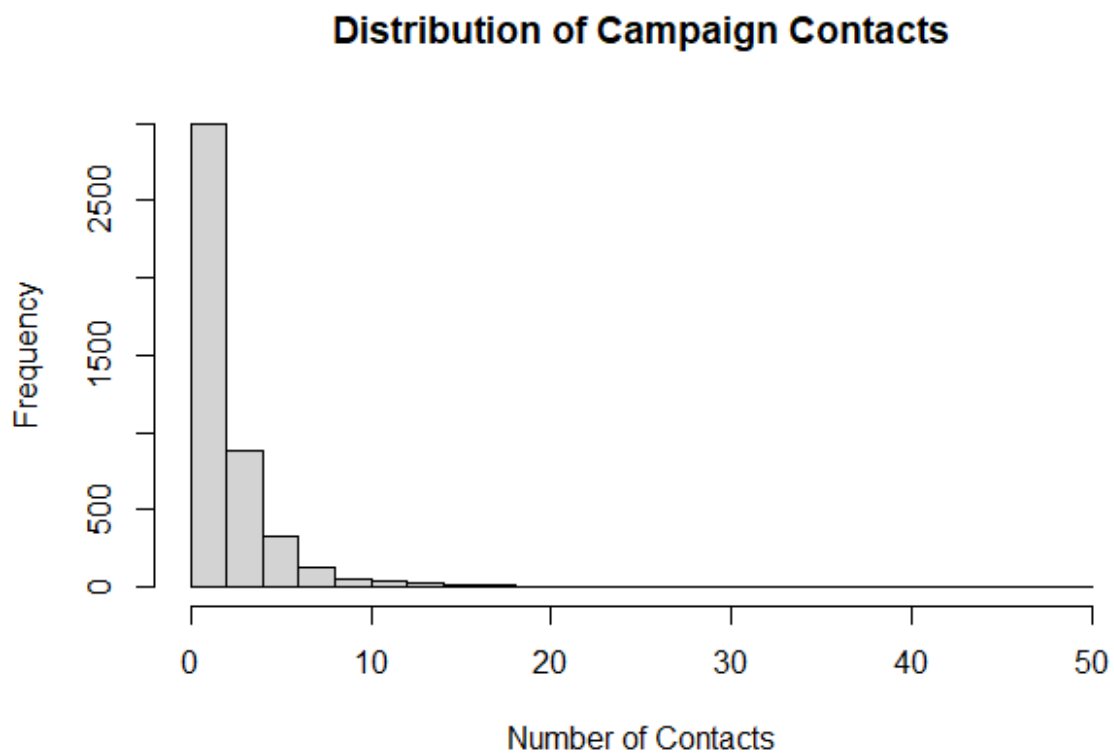


Figure 14: Distribution of Campaign Contacts

This distribution of the number of times a customer is contacted follows a strong right-hairy distribution, that is, the overwhelming majority of customers is contacted only a few times.

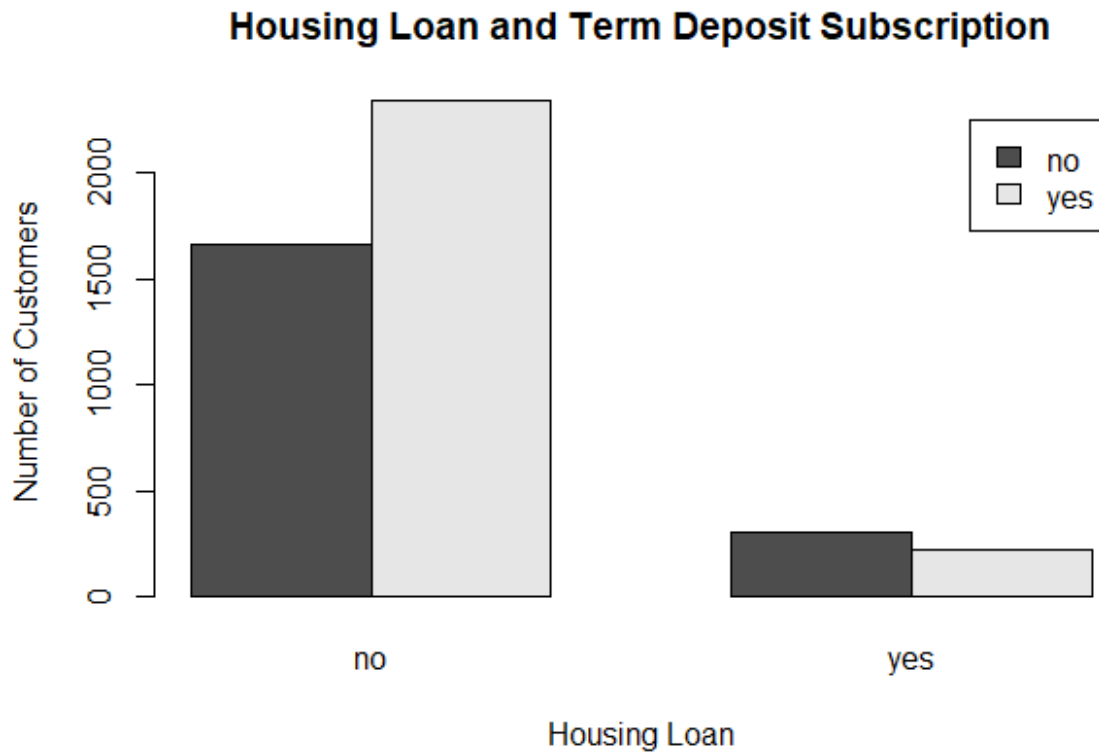


Figure 15: Housing Loan and Term Deposit Subscription

Customers with no housing loans display relatively more successful subscriptions to the size of the group than customers with housing loans, implying a potential link that needs formal statistical testing of hypothesis.

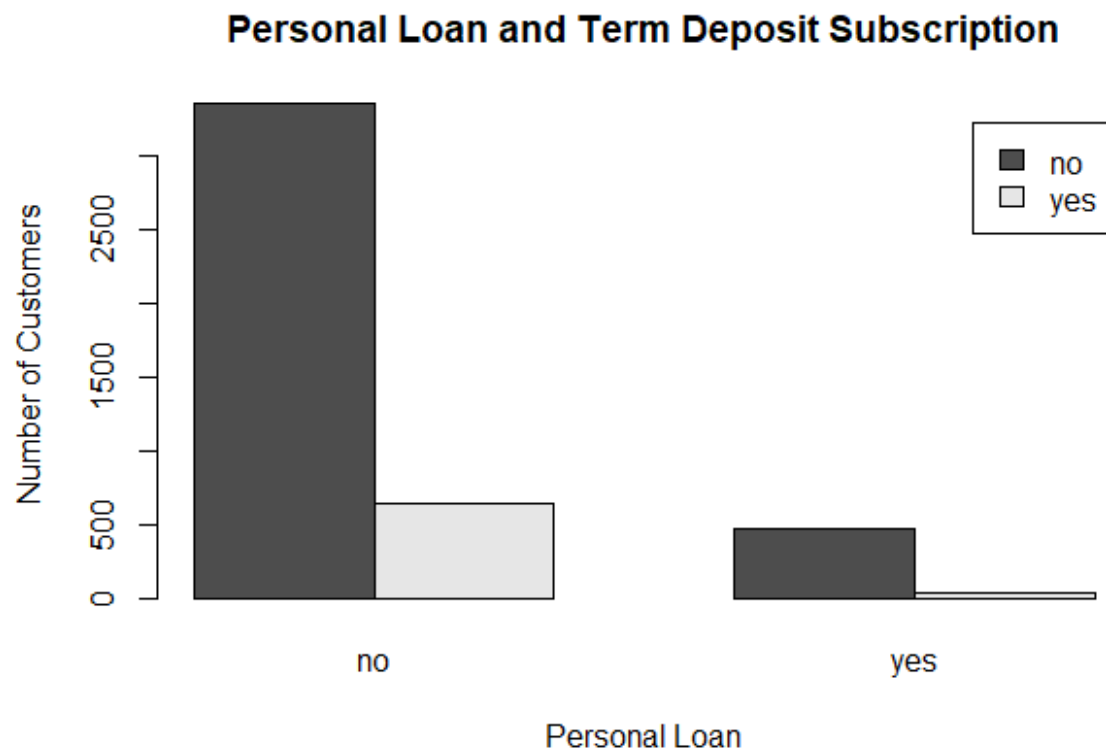


Figure 16: Personal Loan and Term Deposit Subscription

The majority of customers had no personal loans and non-subscription was still predominant at both categories.

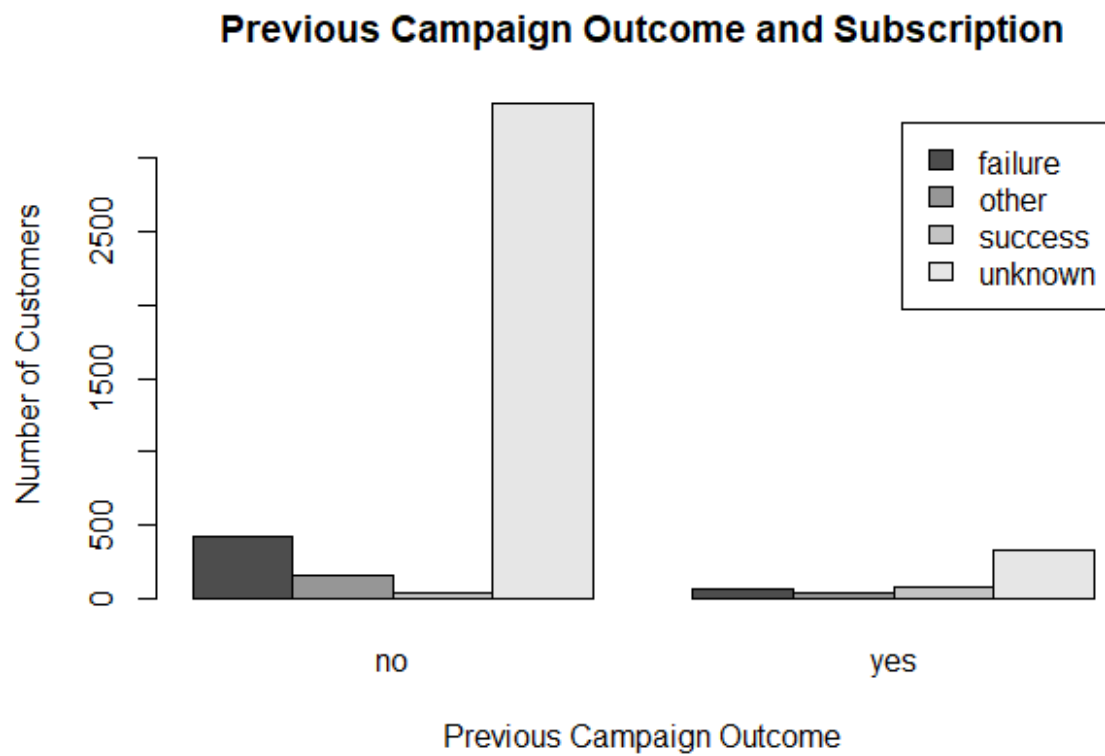


Figure 17: Previous Campaign Outcome and Term Deposit Subscription

The observations are classified into "The unknown", and previous campaign success is relatively more common among those who subscribe, indicating that successful previous campaigns might be linked to future subscription behaviour.

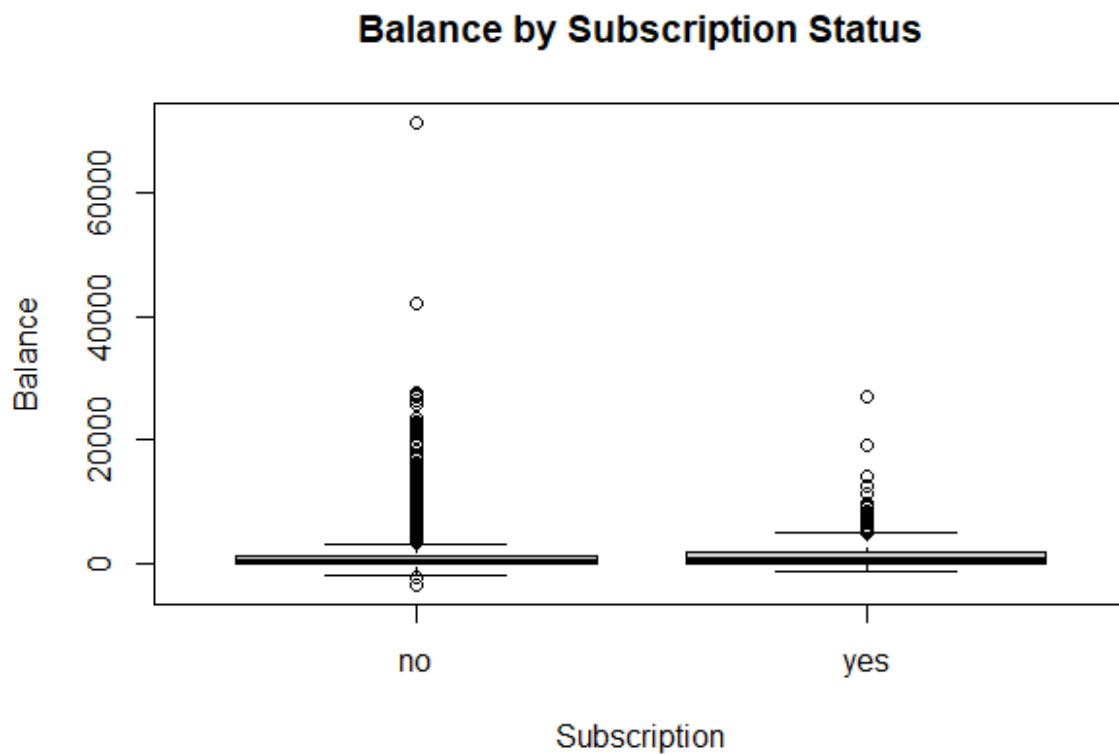


Figure 18: Customer Balance by Subscription Status

There is a significant variation with millions of extreme 'outliers' in account balances, especially for non-subscribers.

4.3 Initial Findings

The exploratory analysis revealed some key patterns in the Bank Marketing data set. The number of non-subscribers to term-deposits was very high, compared to that of subscribers (4 000 to 521). There was a higher focus on the age of the customers, primarily in the 30 to 60 years age group with the age spread able to occur almost evenly in each subscription. The distribution of the accounts was highly right-skewed and had a number of extreme values (Hamedani *et al.*, 2023). Call duration also exhibited a positive skewness distribution and subscribers had longer calls, generally, than non-subscribers (Miller *et al.*, 2025). During the campaign most of the customers were only contacted few times. Those who did not have a house or personal loans, recorded a higher percentage of subscriptions. These patterns indicate that factors related to finances and campaign should be further analyzed by way of inferential statistical tests.

5. Hypothesis Development

Four hypotheses were formulated based on the exploratory analysis to examine the relationship between customer characteristics, campaign factor and term-deposit subscription. A 5% significance level ($\alpha = 0.05$) was used. The first three hypotheses are categorical and analyzed using the chi-square tests; call duration is compared between the groups in subscription using the Wilcoxon rank-sum test.

5.1 Hypothesis 1: Housing Loan and Subscription

H0: Housing loan status and term-deposit subscription are independent; there is no significant association between them.

H1: Housing loan status and term-deposit subscription are associated; there is a significant association between them.

5.2 Hypothesis 2: Personal Loan and Subscription

H0: Personal loan status and term-deposit subscription are independent; there is no significant association between them.

H1: Personal loan status and term-deposit subscription are associated; there is a significant association between them.

5.3 Hypothesis 3: Previous Campaign Outcome and Subscription

H0: Previous campaign outcome and current term-deposit subscription are independent; there is no significant association between them.

H1: Previous campaign outcome and current term-deposit subscription are associated; there is a significant association between them.

5.4 Hypothesis 4: Call Duration and Subscription

H0: The distribution of call duration is the same for customers who subscribed and those who did not subscribe.

H1: The distribution of call duration differs between customers who subscribed and those who did not subscribe.

These hypotheses directly relate to the assessment requirement which is to form testable hypotheses prior to choosing and performing appropriate inferential statistics.

6. Inferential Statistical Method

6.1 Selection of Statistical Test

Four inferential tests were chosen based on the variables and research questions. The relationship between the two variables (housing-loan status and subscription) and the previous campaign outcome and subscription were examined using Pearson's Chi-square test. An

analysis of variance (ANOVA) was applied to compare the mean customer age, whether or not it was different between education group(s). As the duration had non-normal characteristics, the distribution of durations was compared between subscribers and non-subscribers using a Wilcoxon rank sum test (Demirel *et al.*, 2024). In addition, Spearman correlation was used to check the correlation between the numerical variables (Muhammad, 2023). A binary logistic regression approach was then used as the target variable (subscription) is dichotomous and multiple characteristics of customers and campaigns were examined simultaneously, facilitating the interpretation of the results via coefficient, odds ratio and confidence interval.

6.2 Assumption Checking

Assumptions were checked before the results of the inferential analysis were interpreted. In Chi-Square Tests, expected frequencies have been used and unless the cells had frequencies lower than 5, they have been checked to ensure they satisfy the “expected-frequency requirement,” and cells have been examined for expected-frequency values to satisfy the expected-frequency requirement. To apply ANOVA, normality of residuals was determined visually with a histogram and a normal plot of residuals as well as statistically with Shapiro-Wilks parametric test and equality of group variances was tested with Bartlett’s test (Purnama, 2023). Duration of the calls was visually inspected with group histograms and Q-Q plots prior to running the non-parametric Wilcoxon test (Blanca *et al.*, 2023). These checks will help ascertain if the statistical approach chosen is suitable and if not other approaches may be suitable if the assumptions are significantly violated.

7. Statistical Analysis and Results

7.1 Application of the Test

The statistical tests chosen were analysed using the R software at 5 % level of significance ($\alpha = 0.05$). Pearson's chi square tests were performed for the Marital status, Housing loan, and Previous campaign result vs Term-deposit subscription. One-Way ANOVA was used to test the difference between means of customers' age for the different level of education followed by Tukey's post hoc test for the group comparison. Call duration distributions were compared for subscribers and non-subscribers, using the Wilcoxon rank-sum test. Selected numerical variables were also correlated by means of Spearman correlation.

7.2 Statistical Results

```

> housing_table <- table(bank$housing, bank$y)
> housing_table

      no  yes
no  1661  301
yes 2339  220
> # Run Pearson Chi-square test
> housing_test <- chisq.test(housing_table)
> housing_test

Pearson's Chi-squared test with Yates' continuity correction

data:  housing_table
X-squared = 48.885, df = 1, p-value = 2.715e-12

> # Check expected frequencies
> housing_test$expected

      no      yes
no 1735.899 226.1009
yes 2264.101 294.8991
> # Check assumption: all expected frequencies should normally be >= 5
> all(housing_test$expected >= 5)
[1] TRUE
> # Examine residuals
> housing_test$residuals

      no      yes
no -1.797689  4.981105
yes  1.574088 -4.361542
> # Subscription percentages within housing groups
> round(prop.table(housing_table, margin = 1) * 100, 2)

      no  yes
no   84.66 15.34
yes  91.40  8.60
> # Effect size: Phi coefficient
> housing_phi <- sqrt(as.numeric(housing_test$statistic) /
+                      sum(housing_table))
> housing_phi
[1] 0.1039846
> # Decision
> housing_p <- housing_test$p.value
> if(housing_p < 0.05) {
+   print("Reject H0: Housing loan status is significantly associated with subscription.")
+ } else {
+   print("Fail to reject H0: There is insufficient evidence of an association.")
+ }
[1] "Reject H0: Housing loan status is significantly associated with subscription."
>

```

Figure 19: Pearson Chi-Square Test – Housing Loan and Subscription

The Chi-square test indicated that there was a significant relationship between the housing-loan status and term-deposit subscription, (footnote: $\chi^2(1) = 48.885$, $p < .001$). The percentages of customers who were subscribers without housing loans and with housing loans were 15.34% and 8.60%, respectively. A relatively weak association between the variables ($\Phi = 0.104$).

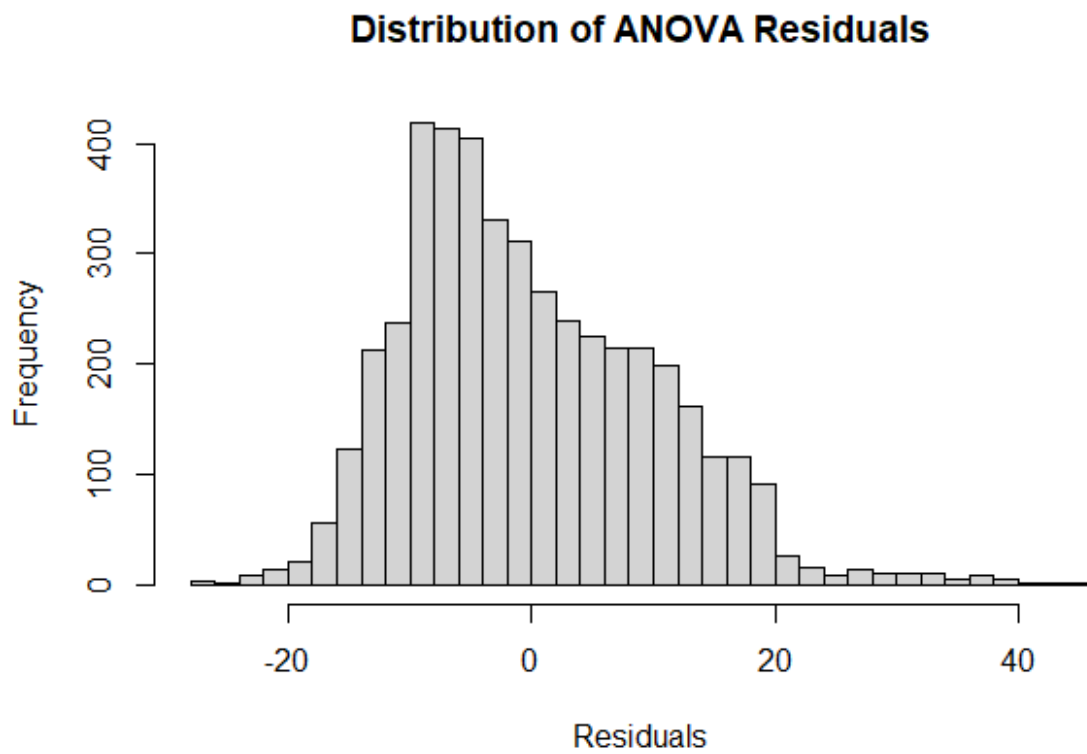


Figure 20: Distribution of ANOVA Residuals

The distribution violates the (close-to-normal) residual distribution expected under the usual one-way ANOVA assumption of normality.

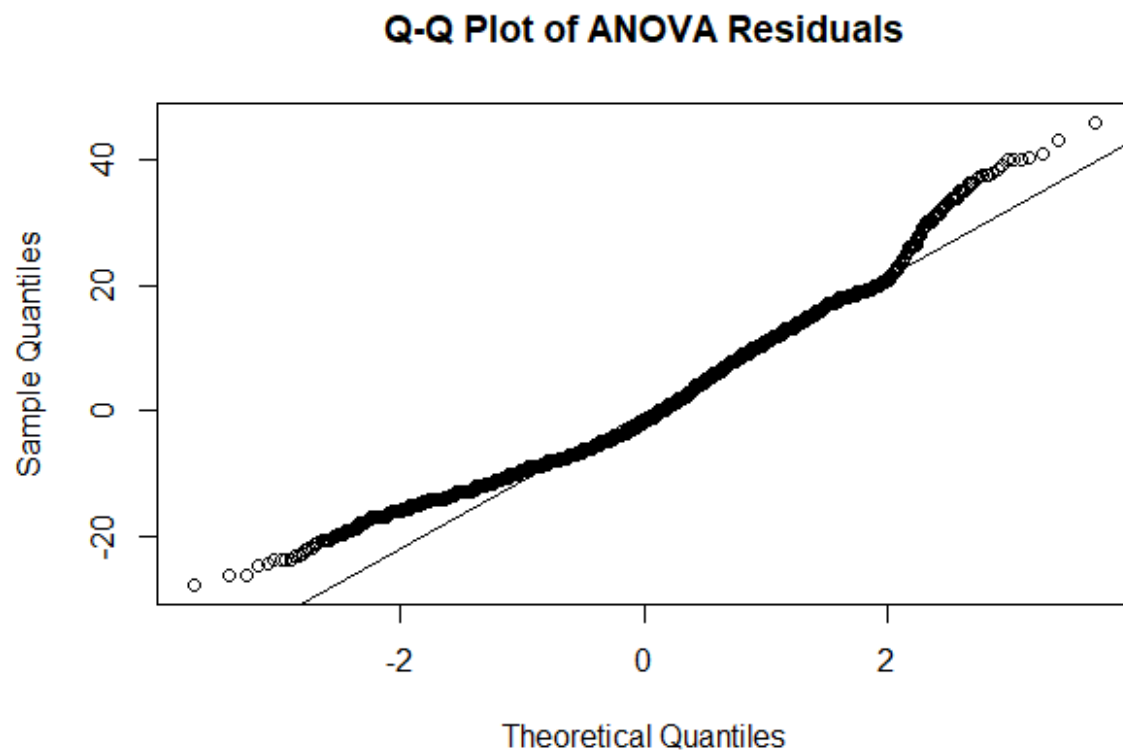


Figure 21: Q-Q Plot of ANOVA Residuals

This shows that the ANOVA residuals are not normally distributed (visual evidence) and caution should be exercised when interpreting the results of standard ANOVA.


```

> shapiro_result

Shapiro-wilk normality test

data: residuals(age_anova)
W = 0.96863, p-value < 2.2e-16

> # CHECK EQUAL VARIANCE ASSUMPTION
> bartlett_result <- bartlett.test(age ~ education, data = bank)
> bartlett_result

Bartlett test of homogeneity of variances

data: age by education
Bartlett's K-squared = 26.281, df = 3, p-value = 8.33e-06

> # EXTRACT ANOVA P-VALUE
> anova_summary <- summary(age_anova)
> anova_p <- anova_summary[[1]][["Pr(>F)"]][1]
> # Display p-value
> anova_p
[1] 1.859102e-61
> # HYPOTHESIS DECISION
> if(anova_p < 0.05) {
+   print("Reject H0 and support H1: At least one education group has a significantly different mean customer age.")
+ } else {
+   print("Fail to reject H0: There is insufficient evidence that mean customer age differs across education groups.")
+ }
[1] "Reject H0 and support H1: At least one education group has a significantly different mean customer age."
> # POST-HOC TEST
> # Run to identify which education groups differ
> TukeyHSD(age_anova)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = age ~ education, data = bank)

$education
      diff      lwr      upr    p adj
secondary-primary -6.7708875 -7.921864 -5.6199106 0.0000000
tertiary-primary  -7.1874074 -8.427528 -5.9472867 0.0000000
unknown-primary   -1.5338681 -3.709994  0.6422583 0.2679475
tertiary-secondary -0.4165199 -1.319375  0.4863357 0.6359838
unknown-secondary  5.2370194  3.233828  7.2402106 0.0000000
unknown-tertiary   5.6535393  3.597834  7.7092451 0.0000000

```

Figure 22: ANOVA Assumption Tests and Tukey Post-Hoc Results

The Shapiro-Wilk test was used to test normality of the residuals, which is rejected at the 0.001-level ($W = 0.96863$, $p < .001$), and the Bartlett test was used to test the variances, which is rejected at the 0.001-level ($\chi^2 = 26.281$, $p < .001$). ANOVA was significant ($p < .001$) but these assumption violations indicate that caution should be used when interpreting the results of ANOVA.

```

R v4.5.1 ~ /
other      159    38
success     46    83
unknown 3368   337
> poutcome_test <- chisq.test(poutcome_table)
> poutcome_test

      Pearson's Chi-squared test

data:  poutcome_table
X-squared = 386.88, df = 3, p-value < 2.2e-16

> # Check expected frequencies
> poutcome_test$expected

      no      yes
failure 433.5324 56.46760
other   174.2977 22.70228
success 114.1340 14.86596
unknown 3278.0358 426.96417
> all(poutcome_test$expected >= 5)
[1] TRUE
> # Examine residuals
> poutcome_test$residuals

      no      yes
failure -0.3137345 0.8693075
other   -1.1587264 3.2106431
success -6.3775893 17.6712669
unknown  1.5713130 -4.3538539
> # Percentages
> round(prop.table(poutcome_table, margin = 1) * 100, 2)

      no  yes
failure 87.14 12.86
other   80.71 19.29
success 35.66 64.34
unknown 90.90  9.10
> # Cramer's V because table is larger than 2x2
> chi_squared <- as.numeric(poutcome_test$statistic)
> n <- sum(poutcome_table)
> r <- nrow(poutcome_table)
> k <- ncol(poutcome_table)
> cramers_v <- sqrt(chi_squared / (n * min(r - 1, k - 1)))
> cramers_v
[1] 0.2925293
> # Decision
> poutcome_p <- poutcome_test$p.value
> if(poutcome_p < 0.05) {
+   print("Reject H0: Previous campaign outcome is significantly associated with subscription.")
+ } else {
+   print("Fail to reject H0: There is insufficient evidence of an association.")
+ }
[1] "Reject H0: Previous campaign outcome is significantly associated with subscription."
>

```

Figure 23: Chi-Square Test – Previous Campaign Outcome and Subscription

There was a significant association between previous campaign outcome and current subscription, $\chi^2(3) = 386.88$, $p < .001$. Only 9.10% for "unknown" outcomes compared to an increase in subscription to 64.34% after the success. Cramer'sV is 0.293 which shows that there is a significant relationship between these variables.

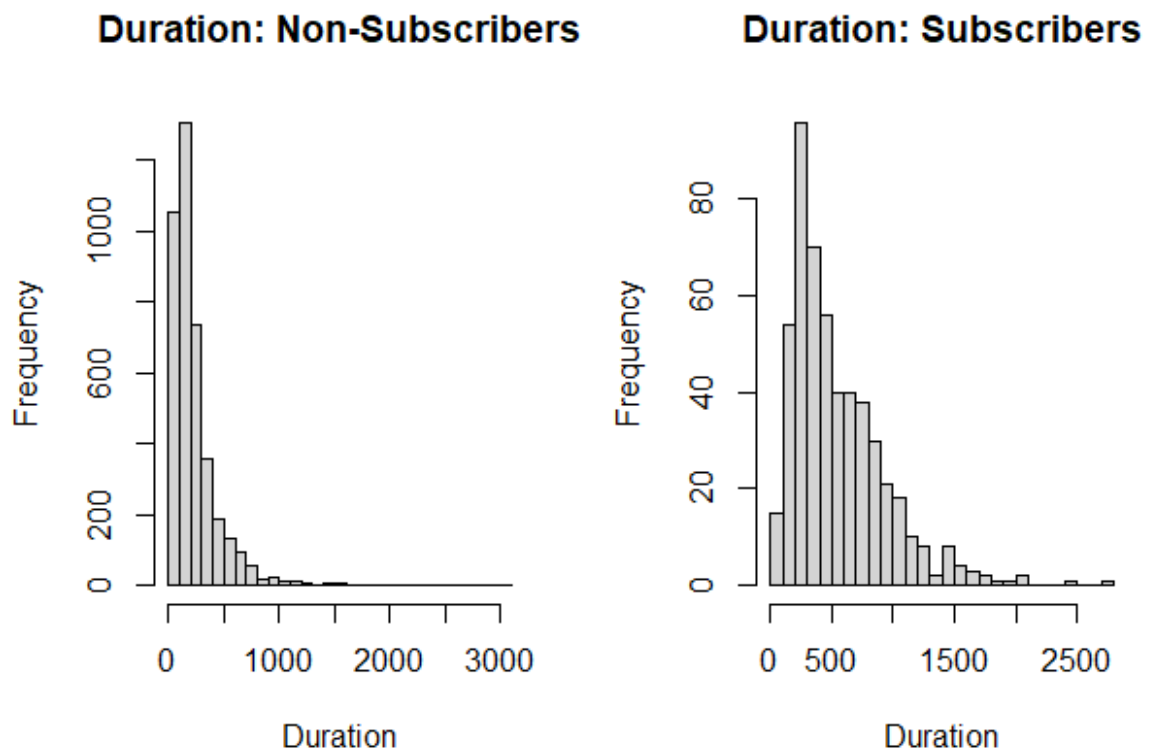


Figure 24: Call Duration Distributions by Subscription Status

The results of the subscribers are more spread out over 2,500 seconds which advocates for a nonparametric comparison rather than a test assuming normal distribution.

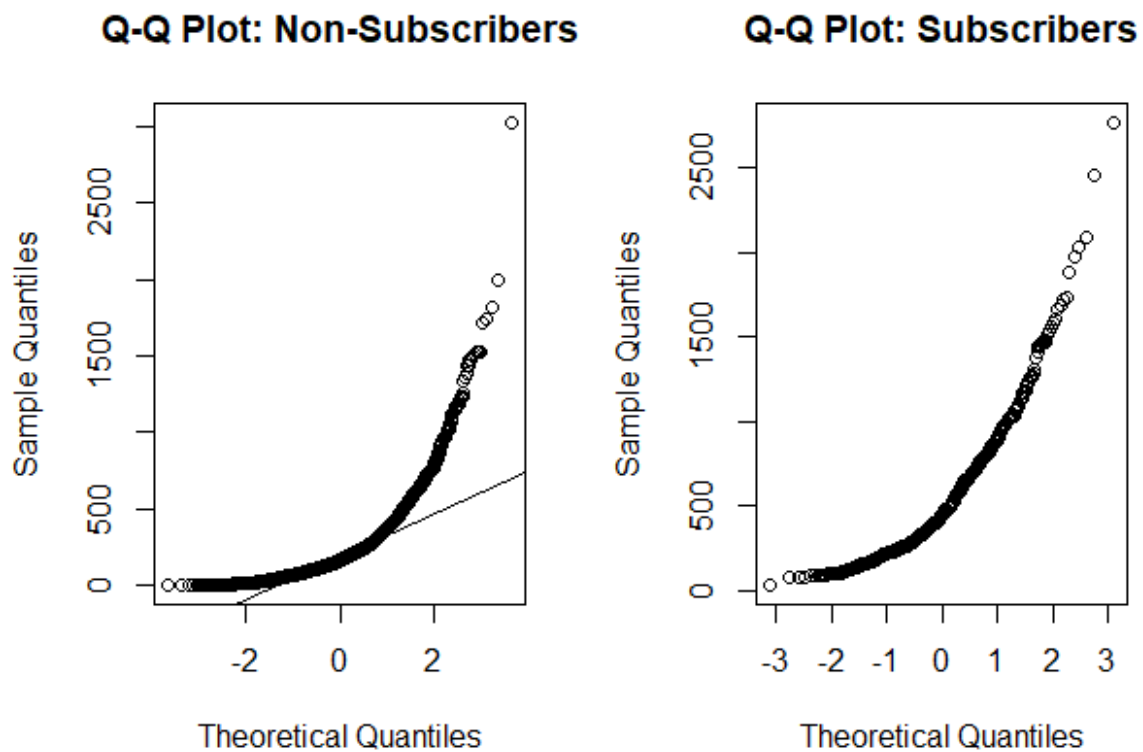


Figure 25: Q-Q Plots of Call Duration by Subscription Status

These results suggest that a non-normality based comparison of two samples (Wilcoxon rank-sum test) be used.

```

-253
> # Decision
> duration_p <- duration_test$p.value
> if(duration_p < 0.05) {
+   print("Reject H0: Call duration differs significantly between subscription groups.")
+ } else {
+   print("Fail to reject H0: There is insufficient evidence of a difference.")
+ }
[1] "Reject H0: Call duration differs significantly between subscription groups."
>

```

Figure 26: Wilcoxon Rank-Sum Test Decision for Call Duration

A Wilcoxon rank-sum analysis of the call-duration distributions showed that the null hypothesis could be rejected in favor of the hypothesis that the call-duration distributions differ markedly between subscribers and non-subscribers. The strongly non-normal distributions noted above, in conjunction with the method being a comparison of two independent groups, indicate that the non-parametric Wilcoxon procedure is an appropriate comparison.

```

> correlation_variables <- bank[, c("age", "balance", "duration",
+                                "campaign", "pdays", "previous")]
> correlation_matrix <- cor(correlation_variables,
+                           use = "complete.obs",
+                           method = "spearman")
> round(correlation_matrix, 3)

```

	age	balance	duration	campaign	pdays	previous
age	1.000	0.076	-0.037	0.025	-0.001	0.001
balance	0.076	1.000	0.054	-0.033	0.062	0.068
duration	-0.037	0.054	1.000	-0.090	0.036	0.040
campaign	0.025	-0.033	-0.090	1.000	-0.138	-0.137
pdays	-0.001	0.062	0.036	-0.138	1.000	0.986
previous	0.001	0.068	0.040	-0.137	0.986	1.000

```

> |

```

Figure 27: Spearman Correlation Matrix of Numerical Variables

Spearman correlation coefficients between most of the numeric variables are weak. The correlation of pdays and previous is however very high, 0.986.

```

> bank$subscribed <- ifelse(bank$y == "yes", 1, 0)
> table(bank$subscribed)

  0    1
4000 521
> # Fit logistic regression
> # Duration is excluded here if the business objective is to identify
> # customers before/during initial targeting because duration is not
> # known before the marketing call has taken place.
> model <- glm(subscribed ~ age + balance + housing + loan +
+             campaign + previous + education + marital +
+             poutcome,
+             data = bank,
+             family = binomial(link = "logit"))
> summary(model)

Call:
glm(formula = subscribed ~ age + balance + housing + loan + campaign +
    previous + education + marital + poutcome, family = binomial(link = "logit"),
    data = bank)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.927e+00  3.826e-01  -5.036 4.75e-07 ***
age           1.401e-02  5.175e-03   2.707 0.006784 **
balance      -3.589e-06  1.622e-05  -0.221 0.824856
housingyes   -5.199e-01  1.030e-01  -5.046 4.52e-07 ***
loanyes      -6.111e-01  1.692e-01  -3.611 0.000305 ***
campaign     -6.904e-02  2.333e-02  -2.959 0.003083 **
previous      1.127e-02  3.259e-02   0.346 0.729400
educationsecondary 1.804e-01  1.602e-01   1.126 0.260007
educationtertiary  4.044e-01  1.672e-01   2.419 0.015571 *
educationunknown -2.447e-01  2.945e-01  -0.831 0.405985
maritalmarried -4.920e-01  1.458e-01  -3.375 0.000738 ***
maritalsingle  -4.148e-02  1.688e-01  -0.246 0.805862
poutcomeother  5.002e-01  2.300e-01   2.175 0.029661 *
poutcomesuccess 2.327e+00  2.350e-01   9.903 < 2e-16 ***
poutcomeunknown -3.456e-01  1.782e-01  -1.940 0.052423 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3231  on 4520  degrees of freedom
Residual deviance: 2898  on 4506  degrees of freedom
AIC: 2928

Number of Fisher Scoring iterations: 5

```

Figure 28: Binary Logistic Regression Model Summary

Logistic regression analysis revealed that age, housing loan, personal loan, campaign contacts, tertiary education, being married, previous campaign “other,” and previous campaign success were statistically significant predictors. The highest positive coefficient was found for previous campaign success ($\beta = 2.327$, $p < .001$).

```

> # Regression coefficients
> coef(model)
              (Intercept)              age              balance              housingyes              loanyes              campaign              previous
educationsecondary      1.401065e-02      educationtertiary      -3.589325e-06      -5.198516e-01      -6.111074e-01      -6.903524e-02      1.127400e-02
      1.804202e-01      4.044484e-01      educationunknown      -2.447192e-01      -4.919868e-01      -4.147780e-02      5.001782e-01      2.327058e+00
      poutcomeunknown
      -3.456045e-01
> # Odds ratios
> odds_ratios <- exp(coef(model))
> odds_ratios
              (Intercept)              age              balance              housingyes              loanyes              campaign              previous
educationsecondary      0.1455968      1.0141093      0.9999964      0.5946088      0.5427495      0.9332938      1.0113378
      1.1977205      1.4984758      0.7829244      0.6114104      0.9593706      1.6490151      10.2477480
      poutcomeunknown
      0.7077924
> # 95% confidence intervals for coefficients
> confidence_intervals <- confint(model)
waiting for profiling to be done...
> confidence_intervals
              2.5 %              97.5 %
(Intercept)      -2.681664e+00      -1.181144e+00
age              3.830071e-03      2.412507e-02
balance          -3.772619e-05      2.596701e-05
housingyes       -7.223343e-01      -3.182569e-01
loanyes          -9.556310e-01      -2.906629e-01
campaign         -1.172558e-01      -2.583010e-02
previous         -5.590283e-02      7.246598e-02
educationsecondary      -1.275600e-01      5.011997e-01
educationtertiary      8.186847e-02      7.382039e-01
educationunknown      -8.458334e-01      3.136389e-01
maritalmarried      -7.732457e-01      -2.011290e-01
maritalsingle     -3.694353e-01      2.927761e-01
poutcomeother     4.334617e-02      9.471285e-01
poutcomesuccess    1.872647e+00      2.795023e+00
poutcomeunknown     -6.896050e-01      9.831370e-03

```

Figure 29: Logistic Regression Coefficients, Odds Ratios and Confidence Intervals

Previously successful campaigns significantly boost subscription likelihood (Odds-ratio: 10.248), whilst housing loan (Odds-ratio: 0.595), personal loan (Odds-ratio: 0.543) and marital status (Odds-ratio: 0.611) have an inhibitory effect on subscription likelihood, provided other model variables are controlled for.

```

> # 95% confidence intervals expressed as odds ratios
> odds_ratio_ci <- exp(confidence_intervals)
> odds_ratio_ci

```

	2.5 %	97.5 %
(Intercept)	0.06844915	0.3069275
age	1.00383742	1.0244184
balance	0.99996227	1.0000260
housingyes	0.48561735	0.7274159
loanyes	0.38456941	0.7477677
campaign	0.88935763	0.9745006
previous	0.94563102	1.0751562
educationsecondary	0.88024059	1.6507004
educationtertiary	1.08531305	2.0921744
educationunknown	0.42919952	1.3683956
maritalmarried	0.46151271	0.8178069
maritalsingle	0.69112452	1.3401427
poutcomeother	1.04429934	2.5782954
poutcomesuccess	6.50549374	16.3630132
poutcomeunknown	0.50177421	1.0098799

```

> # Combine results
> model_results <- data.frame(
+   Coefficient = coef(model),
+   Odds_Ratio = exp(coef(model)),
+   CI_Lower = exp(confint(model)[, 1]),
+   CI_Upper = exp(confint(model)[, 2])
+ )

```

waiting for profiling to be done...
waiting for profiling to be done...

```

> round(model_results, 3)

```

	Coefficient	Odds_Ratio	CI_Lower	CI_Upper
(Intercept)	-1.927	0.146	0.068	0.307
age	0.014	1.014	1.004	1.024
balance	0.000	1.000	1.000	1.000
housingyes	-0.520	0.595	0.486	0.727
loanyes	-0.611	0.543	0.385	0.748
campaign	-0.069	0.933	0.889	0.975
previous	0.011	1.011	0.946	1.075
educationsecondary	0.180	1.198	0.880	1.651
educationtertiary	0.404	1.498	1.085	2.092
educationunknown	-0.245	0.783	0.429	1.368
maritalmarried	-0.492	0.611	0.462	0.818
maritalsingle	-0.041	0.959	0.691	1.340
poutcomeother	0.500	1.649	1.044	2.578
poutcomesuccess	2.327	10.248	6.505	16.363
poutcomeunknown	-0.346	0.708	0.502	1.010

```

> |

```

Figure 30: Logistic Regression Odds Ratios and 95% Confidence Intervals

The 95% C.I. for previous campaign success is (6.505, 16.363) which indicates that there is significant positive relationship between previous campaign success and subscription. Confidence intervals are still less than one for both housing and personal-loans. Age has OR = 1.014 with a 95% CI of 1.004–1.024.

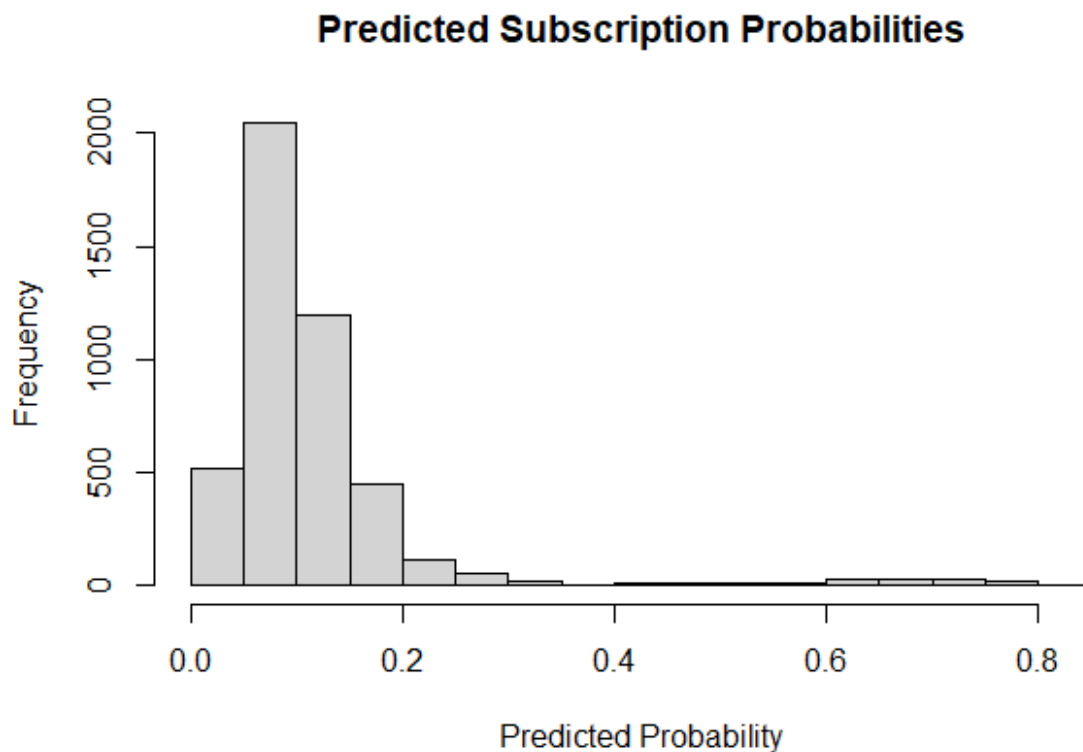


Figure 31: Distribution of Predicted Subscription Probabilities

The distribution of predicted result probabilities of being subscribers also indicates that the majority of these are below 0.20, indicating that the rate of subscription is relatively low. A limited number of customers have a probability greater than 0.5 or 80% and predictions can go as high as 0.8.

```
> hist(bank$predicted_probability,
+       breaks = 30,
+       main = "Predicted subscription Probabilities",
+       xlab = "Predicted Probability")
> # Initial classification threshold of 0.50
> bank$predicted_class <- ifelse(bank$predicted_probability >= 0.50,
+                                1, 0)
> # Confusion matrix
> confusion_matrix <- table(
+   Actual = bank$subscribed,
+   Predicted = bank$predicted_class
+ )
> confusion_matrix
```

	Predicted	
Actual	0	1
0	3960	40
1	444	77

Figure 32: Logistic Regression Confusion Matrix

Neither the positive-class detection is good because it incorrectly marked 40 non-subscribers as subscribers, nor is the negative-class detection because it failed to mark 444 real subscribers.

```

+ )
> performance$value <- round(performance$value, 3)
> performance
      Measure Value
1   Accuracy 0.893
2 Sensitivity 0.148
3 Specificity 0.990
4   Precision 0.658
~ |

```

Figure 33: Logistic Regression Model Performance Measures

The accuracy, specificity, sensitivity and precision of the model were 89.3%, 99.0%, 14.8% and 65.8%, respectively.

```

> # Compare model accuracy with majority-class baseline
> majority_accuracy <- max(prop.table(table(bank$y)))
> majority_accuracy
[1] 0.88476
> cat("Housing Loan Chi-square p-value:",
+     housing_test$p.value, "\n")
Housing Loan Chi-square p-value: 2.7147e-12
> cat("Housing Loan Phi:",
+     housing_phi, "\n")
Housing Loan Phi: 0.1039846
> cat("Personal Loan Chi-square p-value:",
+     loan_test$p.value, "\n")
Personal Loan Chi-square p-value: 2.914829e-06
> cat("Personal Loan Phi:",
+     loan_phi, "\n")
Personal Loan Phi: 0.0695545
> cat("Previous Campaign Chi-square p-value:",
+     poutcome_test$p.value, "\n")
Previous Campaign Chi-square p-value: 1.539883e-83
> cat("Previous Campaign Cramer's V:",
+     cramers_v, "\n")
Previous Campaign Cramer's V: 0.2925293
> cat("Call Duration wilcoxon p-value:",
+     duration_test$p.value, "\n")
Call Duration wilcoxon p-value: 2.329297e-121
> cat("Logistic Regression AIC:",
+     AIC(model), "\n")
Logistic Regression AIC: 2927.957
> cat("Classification Accuracy:",
+     accuracy, "\n")
Classification Accuracy: 0.892944
> cat("Sensitivity:",
+     sensitivity, "\n")
Sensitivity: 0.1477927
> cat("Specificity:",
+     specificity, "\n")
Specificity: 0.99
> cat("Precision:",
+     precision, "\n")
Precision: 0.6581197
> |

```

Figure 34: Final Statistical Results and Model Performance Summary

Housing ($p < .001$, $\Phi = .104$), personal loan ($p < .001$, $\Phi = .070$), previous campaign outcome ($p < .001$, $V = .293$), and call duration ($p < .001$) were significant. Only just over the majority baseline accuracy of 88.5%, model accuracy was 89.3%.

7.3 Hypothesis Decision

The null hypotheses for housing loan, personal loan, the outcome of the previous campaign and call duration were rejected since their corresponding p value was below 0.05 at the 5% level of significance. Hence, these variables were statistically correlated/significant with subscription. The ANOVA results were significant between the education groups as well.

8. Discussion

The analysis revealed that there were significant relationships between the customer financial characteristics, customer campaign history and the subscription of term-deposit. Subscriptions were higher for customers who had no housing or personal loans, but the effect size was quite small. There was a stronger association in the previous campaign outcome—with the likelihood of a subscription being much higher for those who had been successful in the previous campaign. The amount of time people spend on the phone also varied greatly from subscriber to non-subscriber (Yu *et al.*, 2022). The logistic regression confirmed that a number of customer and campaign factors are significant in predicting likelihood of being subscribed. But with a 14.8% sensitivity the model lacked in identifying actual subscribers, even though it had a comparative accuracy of 89.3% and specificity of 99.0%. This was due to the presence of class imbalance.

9. Business Recommendations

The bank should focus on customers that have a set of attributes that indicates they may have greater likelihood of subscribing, especially where they have had a successful campaign outcome. There might also be scope to target the campaign to individuals who do not have a house nor any personal loans (Borges, 2023). Careful approach to repeated contacts. Models can be used to assist targeting; however, the sensitivities are rather low and the model needs to be enhanced for operational use.

10. Limitations and Future Work

10.1 Limitations

There is a large degree of class imbalance in the dataset with 521 subscribers and 4,000 non-subscribers. Categorical (“known”) values are given to a number of variables, in particular the variable on previous campaigns outcome. Also, the assumptions of ANOVA were violated. Moreover, accounting for sensitivity, the logistic model's accuracy was 14.8% which meant that there were many subscribers classified as non-subscribers even though the correctness of the whole model was high.

10.2 Future Work

Research and future work should handle class imbalance and be able to test different classifications to boost the detection of subscribers. A number of other classifications could be used to model performance along with ROC-AUC and cross validation. Non-parametric procedures may enhance analyses that might fail to meet the ANOVA assumptions. Future analyses might also investigate the interplay of the demographic, financial and campaign-related factors.

11. Conclusion

The results confirm the relationship between some financial features of the customers and their subscription to the term deposits, as well as between campaign parameters and subscription to term deposits. The scale items 'housing loans' and 'personal loans' as well as 'call length' and 'previous campaign results' had statistically significant results. Logistic regression was also used to gain further insight into explanation but limited to not being able to produce sensitivity that enables prediction. Overall the results give time to prove evidence on how to conduct future marketing analysis and targeting.

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Appendix:

GitHub	Link	: https://github.com/dhanashhari-commits/B105-Applied-Statistical-Modelling.git
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